ITWP Final Project – Pilvi

Conceptualization

The idea for Pilvi was simple and straightforward – a compact weather application that runs smoothly and has a modern look and feel. Inspiration was heavily drawn from the UI of AccuWeather (It is personally my main app for weather-related information, and is one I use regularly.) Noting the straightforward nature of the app, the 24-hour forecasts and 7-day forecast sections of Pilvi were designed in a manner similar to that of AccuWeather – horizontal scrolling for 24-hour forecast, and vertical list for 7-day forecast. Additionally, it verifies data across three different sources, something that most weather apps either don't do, or if they do, it is not shown to the user.

Tools used

The tools used for this project were APIs, such as:

1. OpenWeatherMap. It is used for current temperature, weather conditions, humidity, wind speed, and forecast comparison data. The authentication method is the API key.
2. Open-Meteo. It is used for hourly temperatures, weather codes, daily highs and lows, and the seven-day forecast. There was no authentication required for basic access, and hence no API key either.
3. Finnish Meteorological Institute (FMI). It is used for temperature forecast data from Finland's meteorological service, intended as the third line in the comparison chart (last section of Pilvi). It also required no authentication.
4. Geolocation. To obtain the present location of the user.

I also used a library for the chart-building, which was Chart.js. It allowed me to create the three-provider temperature comparison graph.

Usage guide

It's simple – go to Pilvi (naadiyasaikia.github.io/Pilvi)! The website is runnable on both mobile and PC. Otherwise, simply download the ZIP file from the GitHub, extract it and run index.html! (This methos only works for PC).

Point allocation

The project should ideally score at least 40/40 points. The following was executed:

Feature	Max points	Implemented?
Well written PDF report	3	Yes
No report	-30	-
Application is responsive and can be used on both desktop and mobile environment	4	Yes
Application is not responsive	-2	-

Application works on Firefox, Safari, Edge and Chrome	3	Yes
The application has clear directory structure and everything is organized well	2	Yes
Application does not work	-30	-
CSS, JavaScript and HTML are all in the same file	-5	Yes
Inappropriate content, including hate speech -related memes and other trash	-100	-
<Your own feature>: Multiple ways to search for a city. You can either hover and click the search button, or you can even click the Enter key on your keyboard. Works on both PC and mobile, across all browsers.	1	**Yes**
<Your own feature>: Horizontally scrollable hourly forecast. It makes the application feel compact, and doesn't overly clutter the screen with symbols from every hour of the day.	1	**Yes**
<Your own feature>: Persistent favorite locations. They are locally stored in the browser, meaning that every time you open the website on the same browser, it remembers favorited locations in every session.	1	**Yes**
<Your own feature>: GitHub hosting. Works on every single device flawlessly, thanks to the repository being hosted as a website on GitHub's domain.	1	**Yes**

Project specific features:

Feature	**Max points**	**Implemented?**
User can search for locations	1	Yes
User can use his/her location GPS-coordinates (Geolocation API)	2	Yes
At least two data/forecast providers are used (this means completely different data sources like x.com and y.com, not just different API endpoints on same service)	3	Yes
At least three data/forecast providers are used	2	Yes

User sees the current weather at a specific location	1	Yes
User sees the forecast for the next 24 hour, hourly based	3	Yes
User sees the forecast for the next 7 days	3	Yes
All the weather forecast elements uses icons (and numbers) for e.g. sunny and cloudy weathers	3	Yes
The look and feel of the application reflects the current weather (e.g. it is blueish, when it is cold; reddish, when it is hot;, dark, when it is night...)	2	Yes
User sees simultaneously two forecast in a graph, e.g. there is temperature forecast for the next 24 hours and there are two lines telling how the data sources are providing (a bit) different data	3	Yes
User has the option to tag some locations as her favorites and thus access them from the favorites menu	2	Yes
User has an option to switch between celsius and fahrenheit degrees and kelvins	2	Yes

AI Declaration

AI was used in fleshing out only the more complex functions (such as chart generation, integration of multiple APIs, and location-favoriting feature). It was also slightly used for some formatting (such as depicting temperature change). The AI agent used was ChatGPT (free model).